Ben Lewis

Senior Software Engineer

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Senior software engineer with 5+ years of experience designing backend-heavy systems for optimisation and scheduling software. My work focuses on domain modelling, internal DSL design, performance optimisation, and integration test architecture, with end-to-end ownership across architecture, implementation, rollout, and iteration.

Education

Bachelor of Mathematics and Bachelor of Engineering (Honours)

Queensland University of Technology

Feb 2015 - Nov 2021

- Major in applied/computation mathematics and computer/software engineering
- Honors thesis topic: Exploring Parallel Periodic Distance Transform Algorithms
- Grade point average of 6.3

Queensland Certificate of Education

Hills International College

Jan 2013 - Dec 2014

- College captain and dux
- Overall position (OP): 2 (ATAR 98.0)

Work Experience

Senior Software Engineer

Deswik

Jan 2021 - Present

- Promoted from Graduate Software Engineer to Software Engineer, then to Senior Software Engineer while contributing across the Optimization (GO, Blend, LHS) and Surface Planning (NOVA) teams
- Designed and delivered a product-agnostic expression language platform for optimisation and planning workflows, replacing fragile legacy implementations with a formally structured pipeline spanning lexing, parsing, analysis, compilation, type inference, linting, and location-aware diagnostics
- Built modern editor capabilities around the expression system, including syntax highlighting, autocomplete, formatting, tooltips, and localisation support for 10+ languages, materially improving usability for engineering users writing production expressions
- Abstracted the expression platform for reuse across products and provided the API design, documentation, and technical guidance that enabled adoption or planned adoption across 3-4 products; Deswik.BOLT fully adopted the design approach and Deswik.Sched is actively evaluating integration
- Led end-to-end delivery of the expression system across architecture, implementation, rollout, support, and iteration, coordinating with multiple engineers and establishing a reusable internal platform capability rather than a one-off feature
- Designed and implemented integrations with Deswik.BOLT and Deswik.GO to enable cost- and constraint-aware destination scheduling within Deswik.NOVA, delivering a core capability required to position the product as an end-to-end mine planning solution
- Solved large-scale interoperability and workflow orchestration challenges by faithfully mapping product configuration semantics into downstream optimisation systems and preparing high-volume haulage and parcel-flow data for external computation
- Optimised parcel-flow field calculations over large multidimensional datasets by modelling field dependencies as a graph, applying topological ordering, parallelising evaluation, reducing unnecessary flow-link-time-step permutations, and caching results based on expression analysis
- Reduced a representative client model’s parcel-flow calculation runtime from more than 1 hour to 11 seconds, a roughly 1000x improvement that made previously impractical workflows operationally viable
- Worked on systems processing mine planning models with inputs frequently exceeding 1 GB and, in some cases, reaching tens of gigabytes, while keeping solve times within user-acceptable ranges for production workflows
- Designed and introduced an integration testing framework and synthetic model generator that made complex end-to-end test setup fast and repeatable, replacing high-friction

boilerplate and removing a major barrier to writing integration tests
- Increased meaningful integration test coverage from roughly 10-20 tests to more than 890 tests in the following year, with contributions from 12+ engineers, significantly strengthening regression protection across the product
- Helped shift team quality practices by making integration tests a standard part of regression fixes, sharply reducing bugs found during once-per-release manual regression testing and freeing testers to expand scenario coverage
- Mentored 5+ engineers over 5 years, led multi-engineer technical initiatives, and regularly influenced architecture and engineering standards across teams through design guidance, system ownership, and technical direction

Sessional Academic

Queensland University of Technology, Faculties of Science and Engineering
Feb 2018 - Jul 2022

- Responsible for designing, creating and marking of assessment
- Catered facilitation of student learning experiences
- Delivery of lectures, tutorials and workshops of first through third year units:
  - EGB111: Foundation of Engineering Design
  - MXB100: Introductory Calculus and Algebra
  - MXB101: Probability and Stochastic Modelling
  - MZB125: Introductory Engineering Mathematics
  - MZB126: Engineering Computation
  - CAB201: Programming Principles
  - CAB202: Microprocessors and Digital Systems
  - CAB302: Software Development

Learning Host Team Lead

Queensland University of Technology, Student Success Group
Feb 2018 - Jul 2019

- Conducted regular team meetings to brief colleagues on frequent changes and facilitate constructive discussions
- Reported to supervisors periodically on team accomplishments, concerns and needs
- Provided training and development to new and ongoing team members

Learning Host

Queensland University of Technology, Student Success Group
Jul 2018 - Nov 2018

- Provide tier 1 support for students seeking administrative support
- Facilitated and coordinated support spaces for students seeking tier 2 support and higher

Barista

McDonald's, McCafé

Dec 2015 - Dec 2017

- Cultivate valuable skills such as teamwork, leadership and working under pressure
- Provide customers with quality customer service by brewing and serving delicious beverages

Private Tutor

Self Employed

Jan 2013 - Nov 2015

- Provide casual feedback and support for student assessments and homework
- Facilitate learning experiences for highschool students struggling with English, Mathematics and Japanese

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Volunteering

Student Executive and Training Facilitator

Queensland University of Technology, STIMulate

Feb 2018 - May 2019

- Prepare and deliver formal training for new peer learning facilitators
- Maintain healthy communication channels between peer learning facilitators and university staff
- Collaborate with executive team members to organize and schedule training, team building and celebratory events

Peer Learning Facilitator

Queensland University of Technology, STIMulate

Feb 2016 - May 2019

- Provide comprehensive learning support to Mathematics, IT and Engineering students
- Assist and coach fellow peer learning facilitators in providing student support through collaborative sessions

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Programming Tools

C# .NET TypeScript React WinForms SQLite

Parquet Power BI Git PowerShell Jenkins Azure DevOps

Bitbucket Jira

Languages

English
Native

Japanese
Limited Working Proficiency

I have chosen to omit references from this rendition of my curriculum vitae to ensure my references are not unduly contacted. I will gladly provide references to recruiters or potential employers upon request.